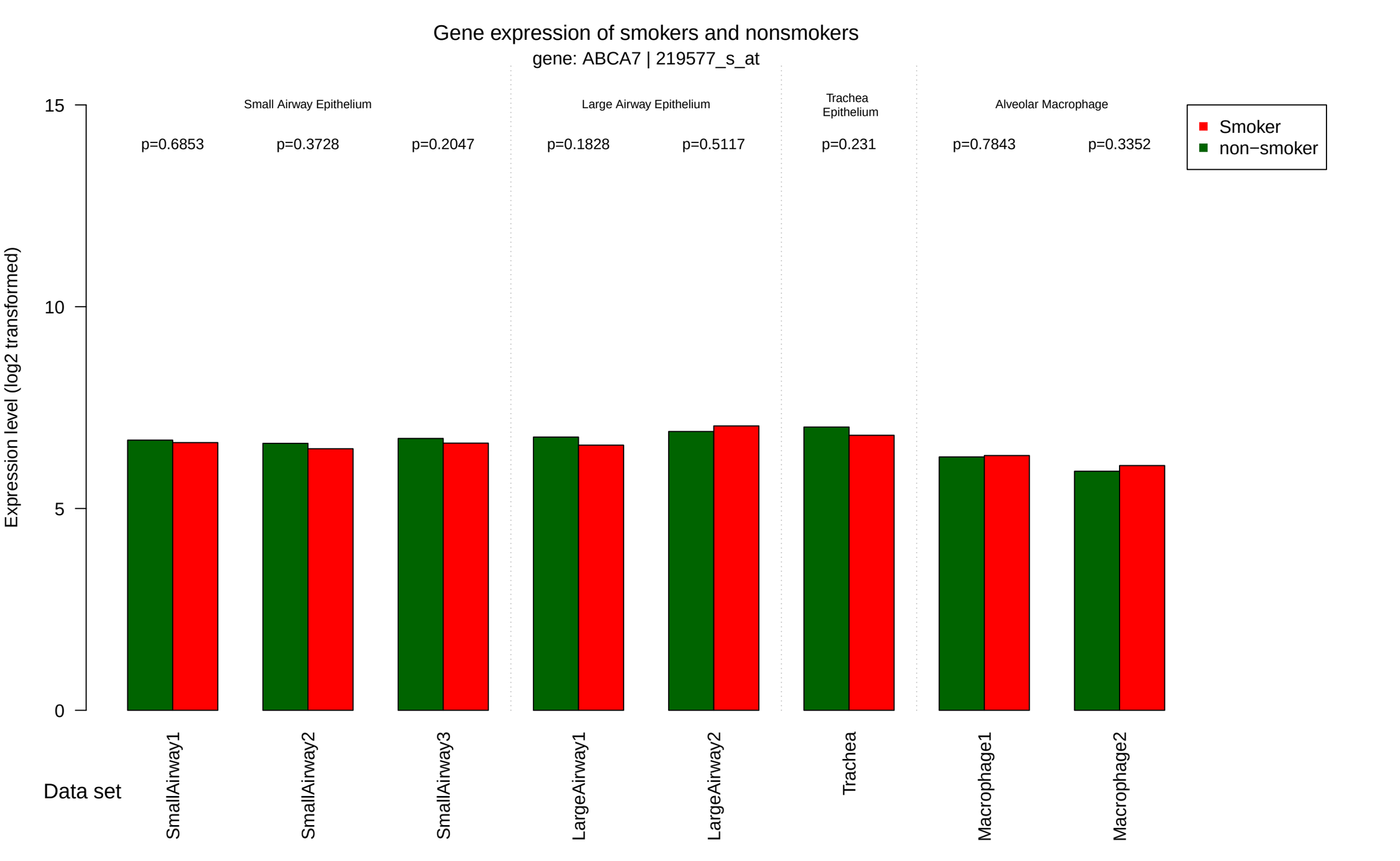
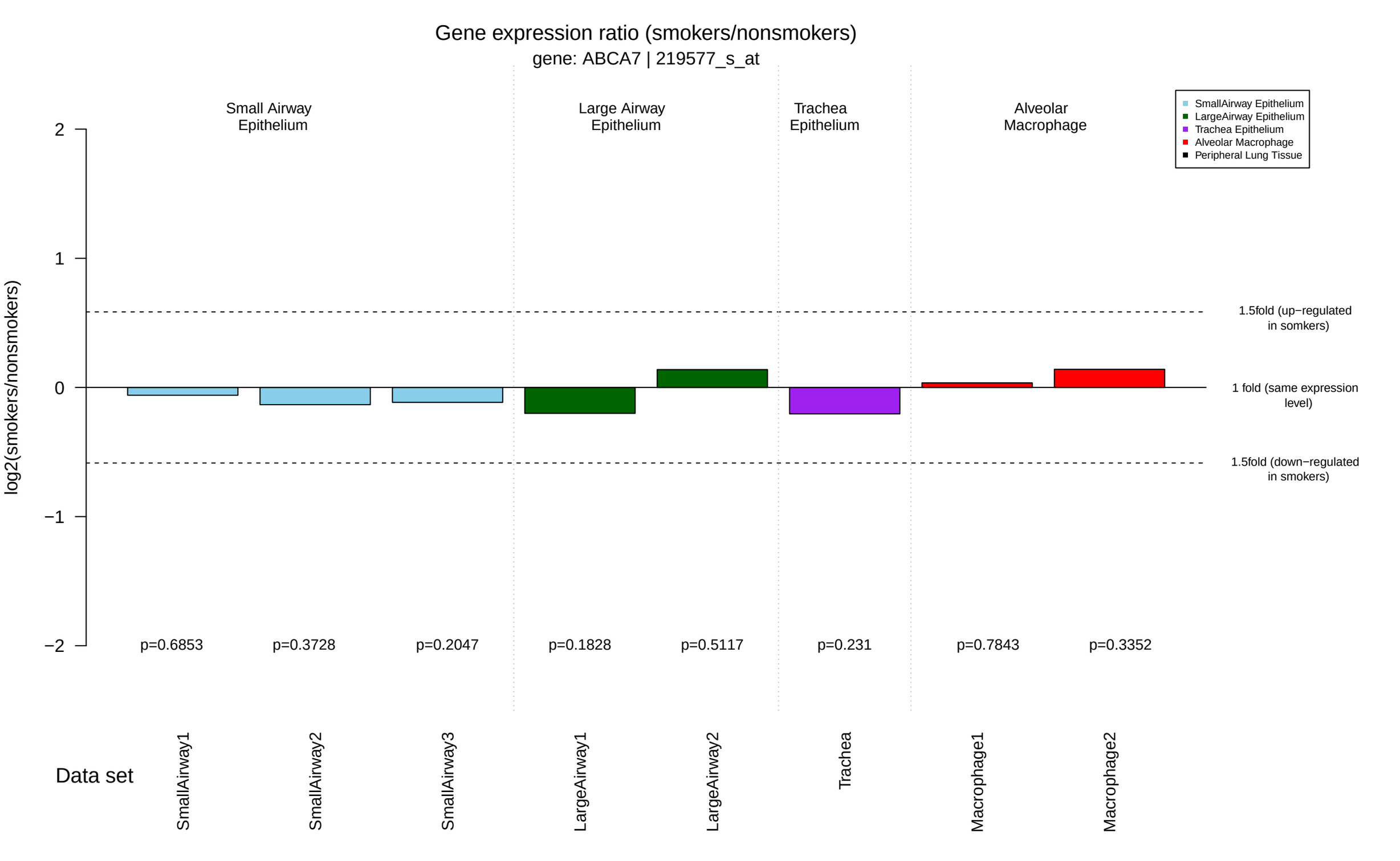
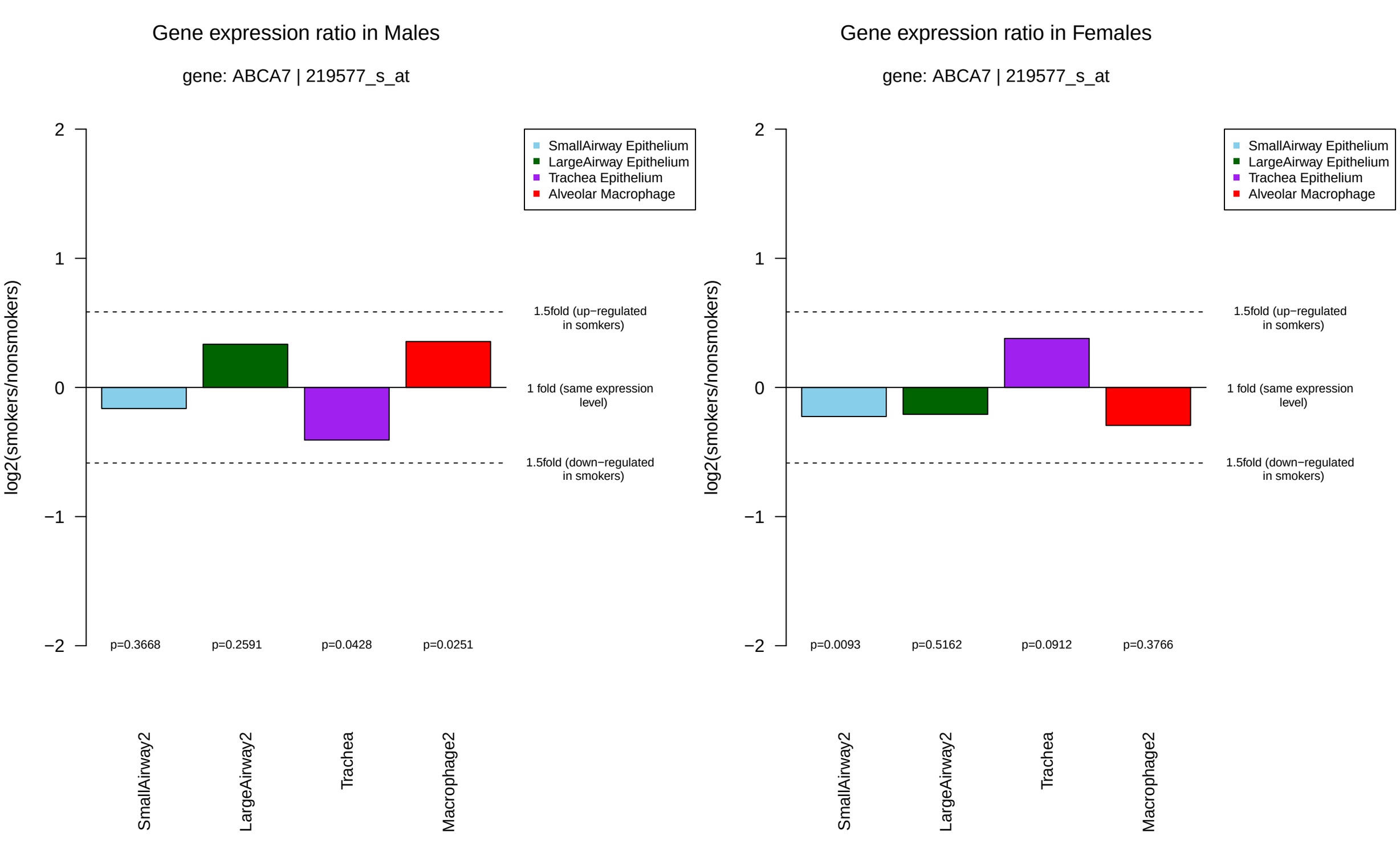
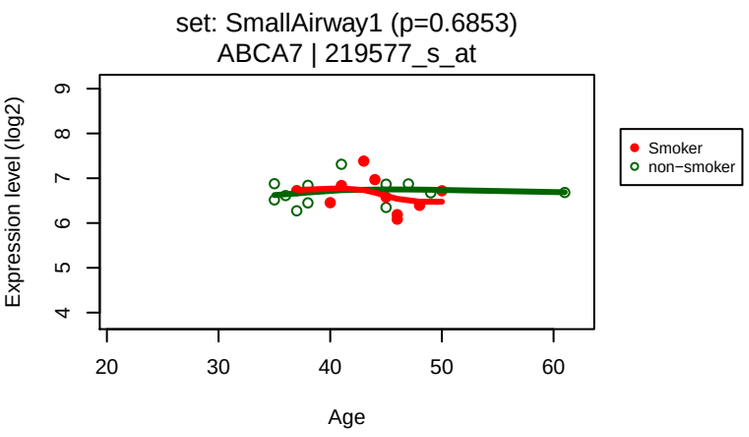
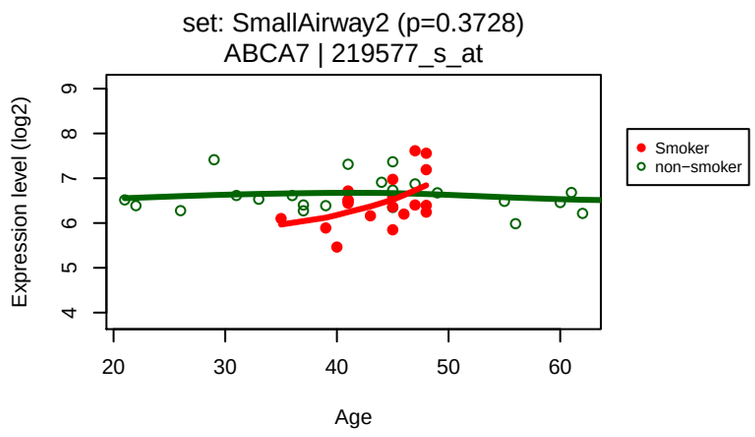
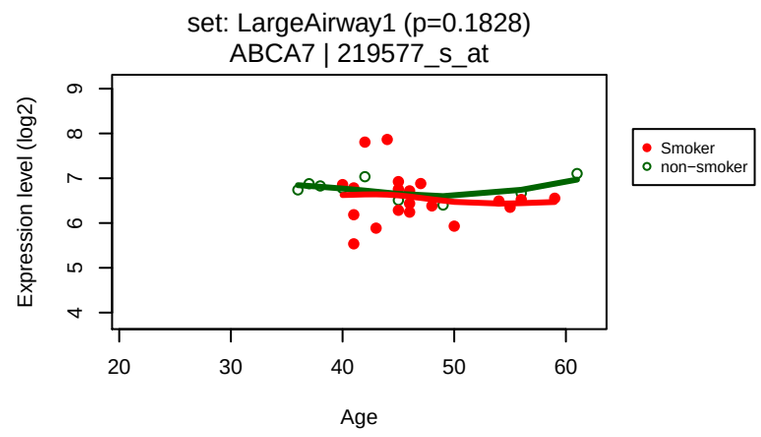

Gene expression of smokers and nonsmokers
gene: ABCA7 | 219577_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6853
p=0.3728
p=0.2047
p=0.1828
p=0.5117
p=0.231
p=0.7843
p=0.3352
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

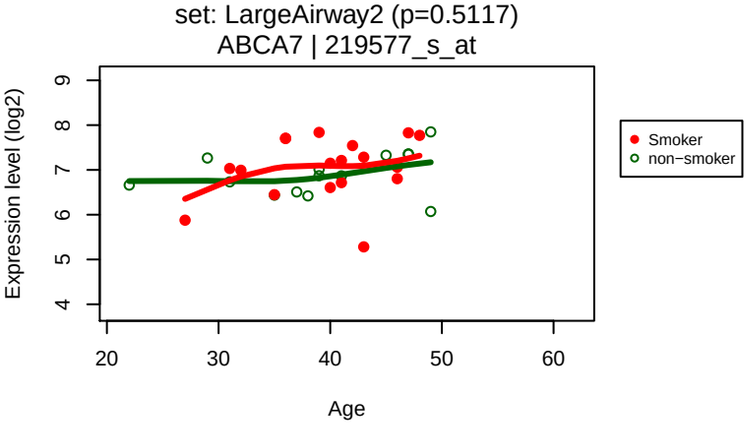

Gene expression ratio (smokers/nonsmokers)
gene: ABCA7 | 219577_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6853
p=0.3728
p=0.2047
p=0.1828
p=0.5117
p=0.231
p=0.7843
p=0.3352
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

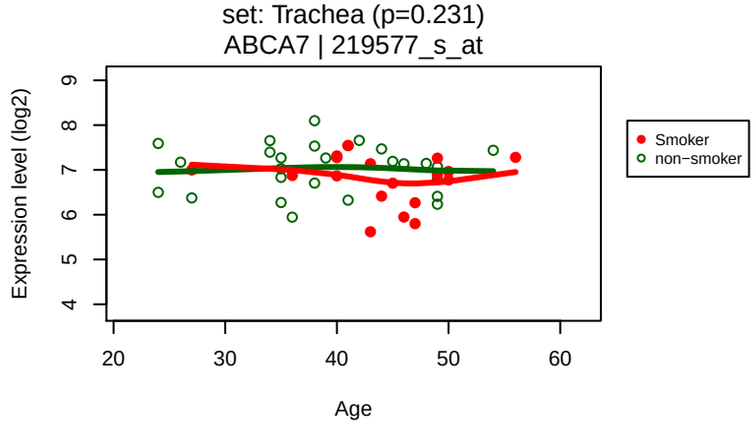

Gene expression ratio in Males
gene: ABCA7 | 219577_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3668
p=0.2591
p=0.0428
p=0.0251
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCA7 | 219577_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0093
p=0.5162
p=0.0912
p=0.3766
SmallAirway2
LargeAirway2
Trachea
Macrophage2

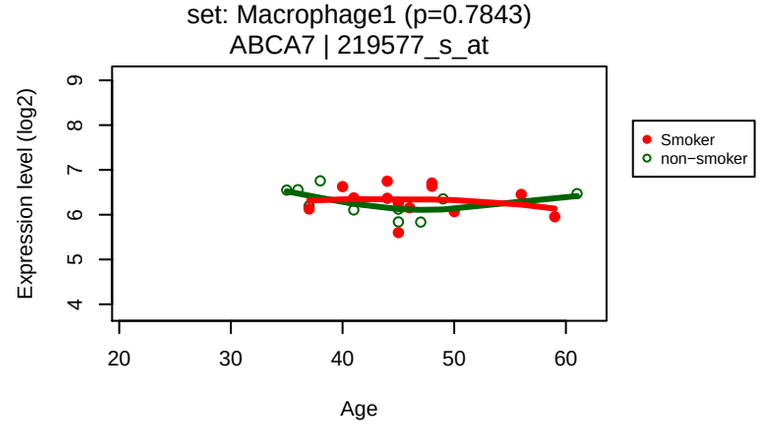

set: SmallAirway1 (p=0.6853)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker

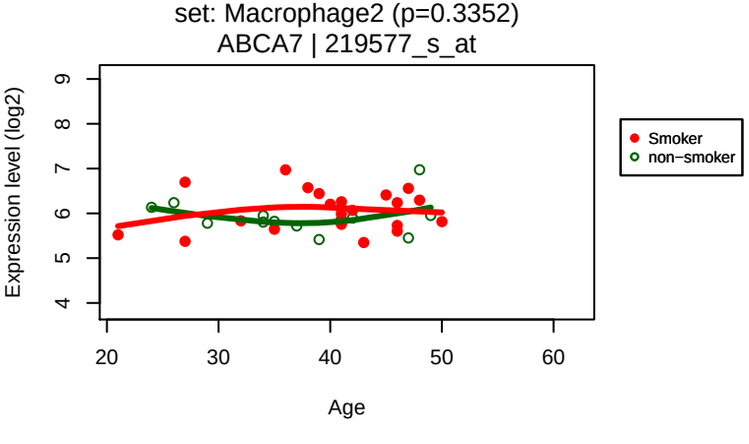

set: SmallAirway2 (p=0.3728)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker

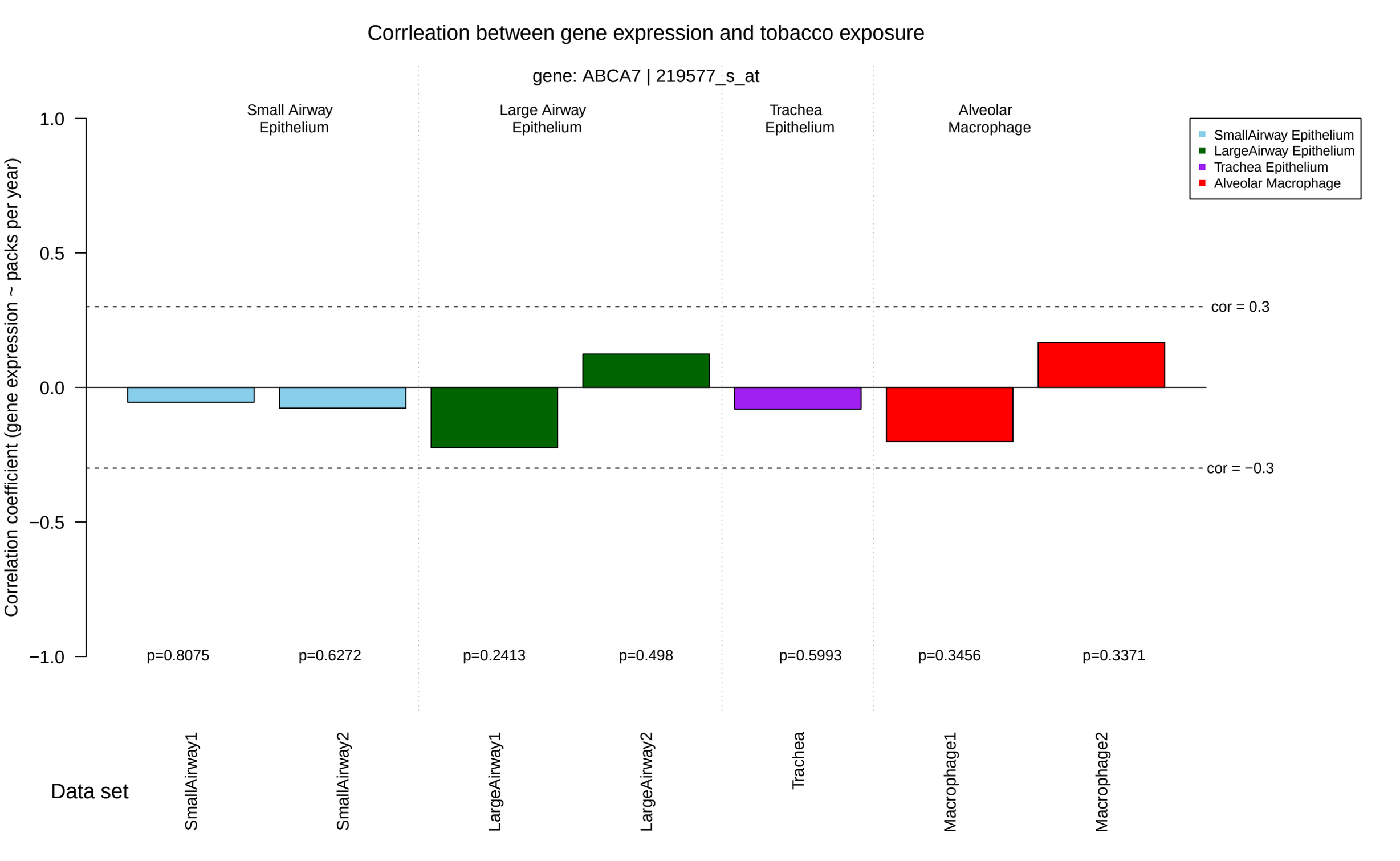

set: LargeAirway1 (p=0.1828)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5117)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.231)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7843)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3352)
ABCA7 | 219577_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCA7 | 219577_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8075
p=0.6272
p=0.2413
p=0.498
p=0.5993
p=0.3456
p=0.3371
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2